

CARBON MONOXIDE**ICSC: 0023 (April 2024)**Carbonic oxide
Carbon oxide**CAS #: 630-08-0****UN #: 1016****EC Number: 211-128-3**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Extremely flammable. Heating will cause rise in pressure with risk of bursting. Gas/air mixtures are explosive.	NO open flames, NO sparks and NO smoking. Closed system, ventilation, explosion-proof electrical equipment and lighting. Use non-sparking handtools.	Shut off supply; if not possible and no risk to surroundings, let the fire burn itself out. In other cases extinguish with carbon dioxide, water spray, powder. In case of fire: keep cylinder cool by spraying with water. Combat fire from a sheltered position.

AVOID ALL CONTACT!			
	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Shortness of breath. Headache. Weakness. Irregular heartbeat. Chest pain. Nausea. Dizziness. Unconsciousness.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Administration of oxygen may be needed. See Notes. Artificial respiration may be needed. Refer immediately for medical attention.
Skin	See Notes.		
Eyes			
Ingestion			

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Evacuate danger area! Consult an expert! Personal protection: self-contained breathing apparatus. Remove all ignition sources. Ventilation.	According to UN GHS Criteria  DANGER Extremely flammable gas Contains gas under pressure; may explode if heated Toxic if inhaled May damage fertility or the unborn child if inhaled Causes damage to blood if inhaled Causes damage to the blood and the central nervous system through prolonged or repeated exposure if inhaled Transportation UN Classification UN Hazard Class: 2.3; UN Subsidiary Risks: 2.1
STORAGE	
Fireproof. Cool. Keep in a well-ventilated room.	
PACKAGING	

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CARBON MONOXIDE**ICSC: 0023****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

ODOURLESS TASTELESS COLOURLESS COMPRESSED GAS.

Physical dangers

The gas mixes well with air, explosive mixtures are easily formed. The gas penetrates easily through walls and ceilings.

Chemical dangers

Reacts with chlorine, fluorine and sulfur. This produces toxic fumes.
Reacts with metal powders and oxidants. This generates fire and explosion hazard.

Formula: CO

Molecular mass: 28.0

Boiling point: -191°C

Melting point: -205°C

Solubility in water, ml/100ml at 20°C: 2.3 (sparingly soluble)

Relative vapour density (air = 1): 0.97

Flash point: Flammable gas

Auto-ignition temperature: 605°C

Explosive limits, vol% in air: 12.5-74.2

Minimum ignition energy: <0.3mJ

EXPOSURE & HEALTH EFFECTS**Routes of exposure**

The substance can be absorbed into the body by inhalation.

Effects of short-term exposure

The substance may cause effects on the blood. This may result in carboxyhaemoglobinemia and cardiac disorders. Medical observation is indicated. Exposure at high levels could cause death.

Inhalation risk

A harmful concentration of this gas in the air will be reached very quickly on loss of containment.

Effects of long-term or repeated exposure

The substance may have effects on the cardiovascular system and central nervous system. This may result in cardiovascular disorders and nervous system impairment. May cause toxicity to human reproduction or development.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 25 ppm as TWA; BEI issued.

EU-OEL: 23 mg/m³, 20 ppm as TWA; 117 mg/m³, 100 ppm as STEL; (binding):

ENVIRONMENT**NOTES**

Specific treatment may be necessary in case of poisoning with this substance.

Oxygen should be administered exclusively by specially trained first aid or medical personnel.

Carbon monoxide may exist in the form of refrigerated liquid capable of causing FROSTBITE on skin contact.

Depending on the degree of exposure, periodic medical examination is suggested.

There is no odour warning even when toxic concentrations are present.

ADDITIONAL INFORMATION**EC Classification**

Symbol: F+, T; R: 12-23-48/23-61; S: 53-45; Note: E

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